Predicting vortex shedding behind a circular cylinder with physics-informed neural networks

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Abstract

Physics-informed neural networks (PINNs) trained without data on the unsteady wake of a circular cylinder tend toward the steady symmetric solution, which satisfies the Navier–Stokes equations exactly. We study this failure at Re = 100 using lattice Boltzmann reference data validated against Qu et al. (St within 0.04%, C_d 3.5% high). Averaged over four start phases and five seeds, a data-free PINN retains 31.1% of the reference wake amplitude. Causal time-marching raises this to 56.4%, but at equal training budget it adds only 7.7 points. Combining marching with random Fourier features and a Reynolds decomposition gives 72.9%, between 70% and 77% at every start phase, at a relative L_2 error of 0.031. This configuration predicts one shedding period from physics alone, but over five periods it sheds 10.4% slow and ends about half a cycle out of phase. Holding the phase requires 350 supervised points per period taken from the predicted span, which turns the task into reconstruction from sparse measurements. The gain from the combined strategies holds for Re = 60–180. A recurrent sequence PINN does not replace marching. For data-driven forecasting, a physics residual makes a Fourier neural operator worse on the validated data; on an earlier, harder dataset it helps the operator at short horizons, but not a U-Net or a convolutional GRU.

1 Introduction

Above Re ≈ 47 the wake of a circular cylinder sheds a Kármán vortex street at a frequency given by the Strouhal number St = fD/U. Predicting this flow with a data-free PINN is difficult for three reasons. The steady symmetric field satisfies the equations and boundary conditions exactly, so it is an easily reached minimum of the residual loss. Networks learn low frequencies first (spectral bias), which works against an oscillatory wake. The initial condition is a single time slice and is weakly weighted against the residual over the space–time domain. Chuang and Barba [1] found that a data-free PINN behaves like a steady solver, and Shah [2], on a confined cylinder benchmark, found that only supervised combinations of training techniques escape this collapse. Causal training, either by time-marching [3] or by causal weighting [4], has been shown to help on one-dimensional problems, but neither of these cylinder studies tests hard time-window marching. Separately, Renn et al. [5] forecast cylinder wakes with a purely data-driven Fourier neural operator (FNO) [6].

The objective of this work is to reproduce the PINN collapse on validated reference data, to identify which training strategy prevents it under a controlled comparison, and to test whether adding physics to an operator forecast improves it. Here, prediction means computing the flow beyond the data the model is given; the reference data are otherwise used only for scoring. The contributions are an unsteady Navier–Stokes operator for the `underPINN` framework [7], a validated lattice Boltzmann dataset, a phase-averaged ablation with equal-budget controls, and a comparison of physics-informed forecasting across three operator backbones.

2 Reference data

The reference wake is computed with a D2Q9 two-relaxation-time lattice Boltzmann solver written in JAX, with half-way bounce-back on the cylinder and free-slip lateral walls. The inlet is $25D$ upstream of the cylinder in a domain $60D$ long. The production run uses $\mathrm{Re} = 100$, $D = 24$ lattice units, $\mathrm{Ma} = 0.087$, a 1440×1200 grid and 2% blockage. It records 641 snapshots over 20 shedding periods on a 241×121 grid ($\Delta x = 0.083D$, $\Delta t^* = 0.19$). Table 1 compares it with Qu et al. [8].

Table 1: Validation at $\mathrm{Re} = 100$ against Qu et al. [8]. Errors are relative to their values $C_d = 1.317$, $\mathrm{St} = 0.1649$ and $C_l' = 0.2224$; the range is their compilation of eight studies.

quantity	present	benchmark range	error
mean drag C_d	1.3634	1.318–1.336	+3.5%
Strouhal number St	0.1648	0.164–0.166	−0.04%
r.m.s. lift C_l'	0.2362	0.222–0.235	+6.2%

2.1 Validation of the reference solution

Two corrections were needed. First, the free-slip walls reflect sound, so the domain behaves as an acoustic duct with a transverse mode at $\mathrm{St}_a = 0.1155$. Without treatment the wake locks onto this mode over long records ($\mathrm{St} = 0.1132$ over 38 periods). An absorbing layer of $8D$ at each wall that relaxes the populations toward the free-stream equilibrium removes it. Blevins [9] showed experimentally that a transverse acoustic wave can shift cylinder shedding to its own frequency. Second, at $D = 32$, moving the inlet from $8D$ to $25D$ upstream lowers C_d by 4.7%, from +7.4% to +2.3% against Qu et al., and takes St from +3.0% to +0.3%; the domain height has a much smaller effect. The grid is converged at $D = 32$; the production value $D = 24$ reads C_d about 1.4% high and is kept so that the snapshot spacing matches PIV-like resolution. None of the PINN or operator results depend on C_d.

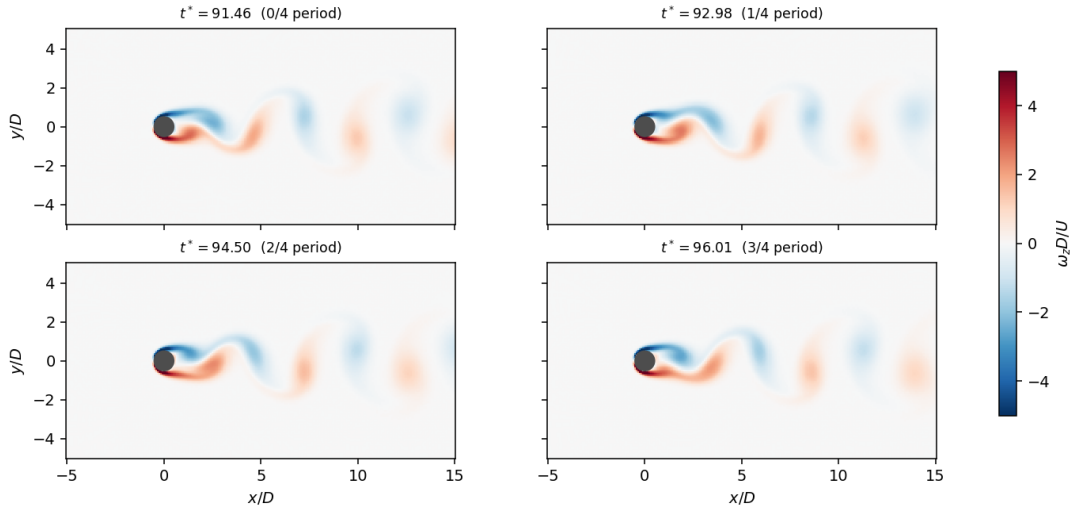


Figure 1: Reference vorticity at quarter-period intervals. These four instants are the four start phases used in Section 4.

3 Methodology

3.1 Unsteady Navier–Stokes operator

The network maps $(x, y, t) \mapsto (u, v, p)$ and is trained on the non-dimensional incompressible equations

$$u_x + v_y = 0, \tag{1}$$

$$u_t + uu_x + vu_y + p_x - \tfrac{1}{\mathrm{Re}}(u_{xx} + u_{yy}) = 0, \tag{2}$$

$$v_t + uv_x + vv_y + p_y - \tfrac{1}{\mathrm{Re}}(v_{xx} + v_{yy}) = 0. \tag{3}$$

`underPINN` provides only a steady residual, so an unsteady operator was implemented, with derivatives computed by forward-mode automatic differentiation. It annihilates the Taylor–Green vortex to 1.6×10^{-16}, matches a reference Hessian implementation to 2.8×10^{-17}, and reduces to the framework's steady residual to 5.6×10^{-17}.

3.2 Training strategies

The baseline is a multilayer perceptron $[3, 128^{\times 6}, 3]$ trained for 20 000 epochs on the equation residual, boundary conditions and initial condition over one shedding period, with no data other than the initial condition. Three strategies are added as independent switches on the same trainer:

- **Fourier features** [10], against spectral bias. The bandwidth is $\sigma = 8$, except in the phase-0 ablation and the supervision study, which use $\sigma = 2$.
- **Reynolds decomposition** $u = u_0 + u'$, where the mean u_0 is a separate frozen network fitted to the measured mean.
- **Causal time-marching**: four windows of a quarter period solved in sequence, each taking its initial condition from the previous window's network. The run uses ground truth only once.

The combination of all three is denoted `all`.

3.3 Evaluation protocol

The main metric is *shedding retained*: the ratio of transverse-velocity r.m.s. at a wake probe between the PINN and the reference. A value near 0% means collapse to steady flow. Field accuracy is the relative L_2 error of the velocity field. Because retention depends on the phase of the cycle at which a run starts, the main comparison is averaged over four start phases with five seeds each. A marched run trains four windows, so each marched configuration is also compared with a single-window run given the same 80 000 epochs.

4 Results

4.1 Baseline PINN

The baseline retains between 19.0% and 43.2% of the wake depending on the start phase, 31.1% on average (Table 2). Training for 80 000 epochs raises the mean to 48.6% but leaves the dependence on start phase in place.

Table 2: Shedding retained by start phase (five seeds each) and relative L_2 error. `all` combines the three strategies.

configuration	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	mean	L_2
baseline	43.2%	20.3%	42.1%	19.0%	31.1%	0.166
marching	70.6%	40.4%	73.8%	40.7%	56.4%	0.156
`all`	74.9%	70.2%	76.6%	70.0%	**72.9%**	**0.031**

4.2 Ablation

In a phase-0 ablation (two seeds), Fourier features (36.9%) and the Reynolds decomposition (36.1%) each retain slightly less than the baseline (41.1%), although the decomposition halves the field error. Marching raises retention at every start phase, from 31.1% to 56.4% on average, but keeps the baseline's dependence on start phase. Only the combination of all three is insensitive to start phase, at a fifth of the field error of either alternative (Table 2, Figure 2). The 95% intervals over seeds are ±1.5, ±0.9 and ±4.6 points for the three rows.

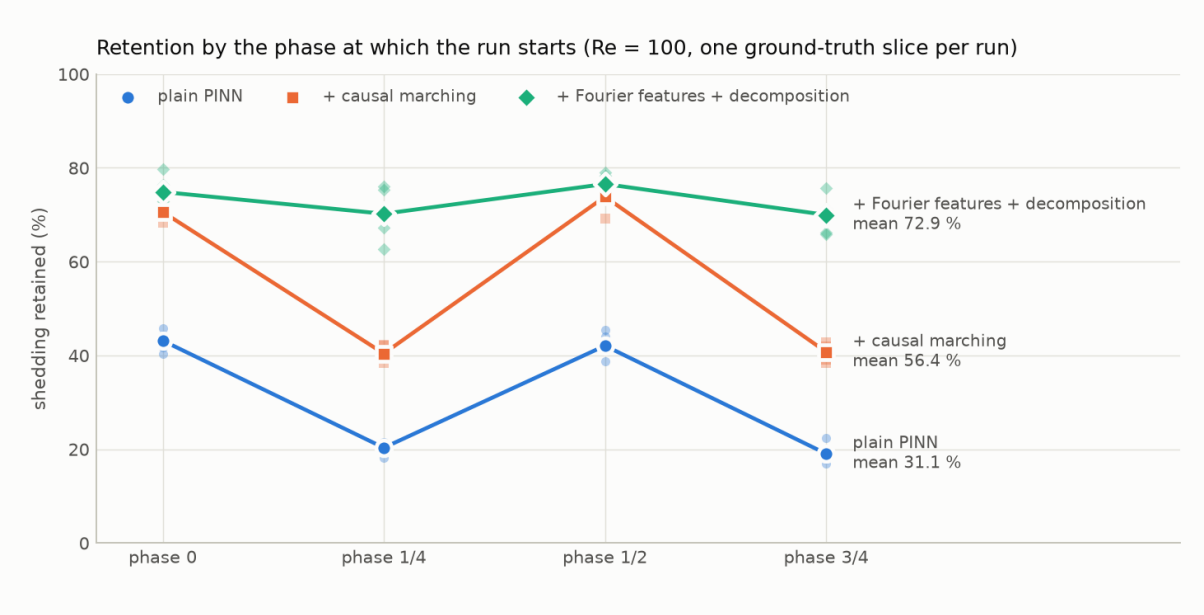


Figure 2: Shedding retained by start phase, all runs.

Table 3 separates the effect of marching from the extra training it receives. Alone, marching beats an equal-budget single window by 7.7 points, less than the 17.5 points that longer training gives the baseline. With Fourier features and the decomposition in place, marching adds 34.1 points and wins all 20 pairings. The strategies are therefore complementary, in agreement with Shah [2]. An earlier version of this study gave each marched window a fresh reference snapshot; that raised retention at phase 0 to 93.0% and 94.5%, so about twenty points of the earlier result came from extra ground truth.

Table 3: Marching against single-window runs of equal budget (80 000 epochs), averaged over four start phases and five seeds. Gains are paired by phase and seed.

comparison	marched	single	gain (points)	paired range
marching vs baseline	56.4%	48.6%	+7.7	−2.2 to +15.8
`all` vs Fourier + decomposition	72.9%	38.7%	+34.1	+20.0 to +55.2

4.3 Prediction horizon and sparse measurements

Over one period `all` matches the reference phase to 0.1°, at 73.5% of its amplitude. Over five chained periods it keeps shedding, but at St = 0.1476, 10.4% below the reference, and ends about half a cycle out of phase (Figure 3); amplitude falls to 63.8% and the field error rises to 0.156. Data-free marching therefore predicts about one period and sustains a wake beyond it, but does not hold the phase.

Measurements from inside the predicted span correct this. With supervised points in every window of `all` (start phase 0, $\sigma = 2$, two seeds), 100 points per window give 87.4% retention and 500 give 96.9%, with the field error falling to 0.010. Table 4 gives selected rows of a sparse re-anchoring study, in which N supervised points are given to every K-th window of the five-period chain and initial conditions stay chained. The phase locks between 300 and 350 points per period: at 350 per period, 1 750 points in total and about 2% of what per-window snapshots would use, both seeds hold the phase to within 0.003 of a period and St to within 0.7%. Because these points lie inside the span being computed, this result is reconstruction from sparse measurements, not forecasting.

Table 4: Sparse re-anchoring of the five-period chain (selected rows, two seeds each). The lag is the cross-correlation lag over the last period, in shedding periods, positive behind the reference.

N	K	points per period	total points	St error (%)	lag
—	—	0	0	−10.4, −10.2	0.369, 0.345
100	2	200	1 000	−4.6, −3.3	0.120, 0.100
150	2	300	1 500	−1.5, −1.3	0.053, 0.042
175	2	350	1 750	−0.2, −0.7	0.002, 0.002
100	1	400	2 000	−0.3, −0.2	0.001, 0.001

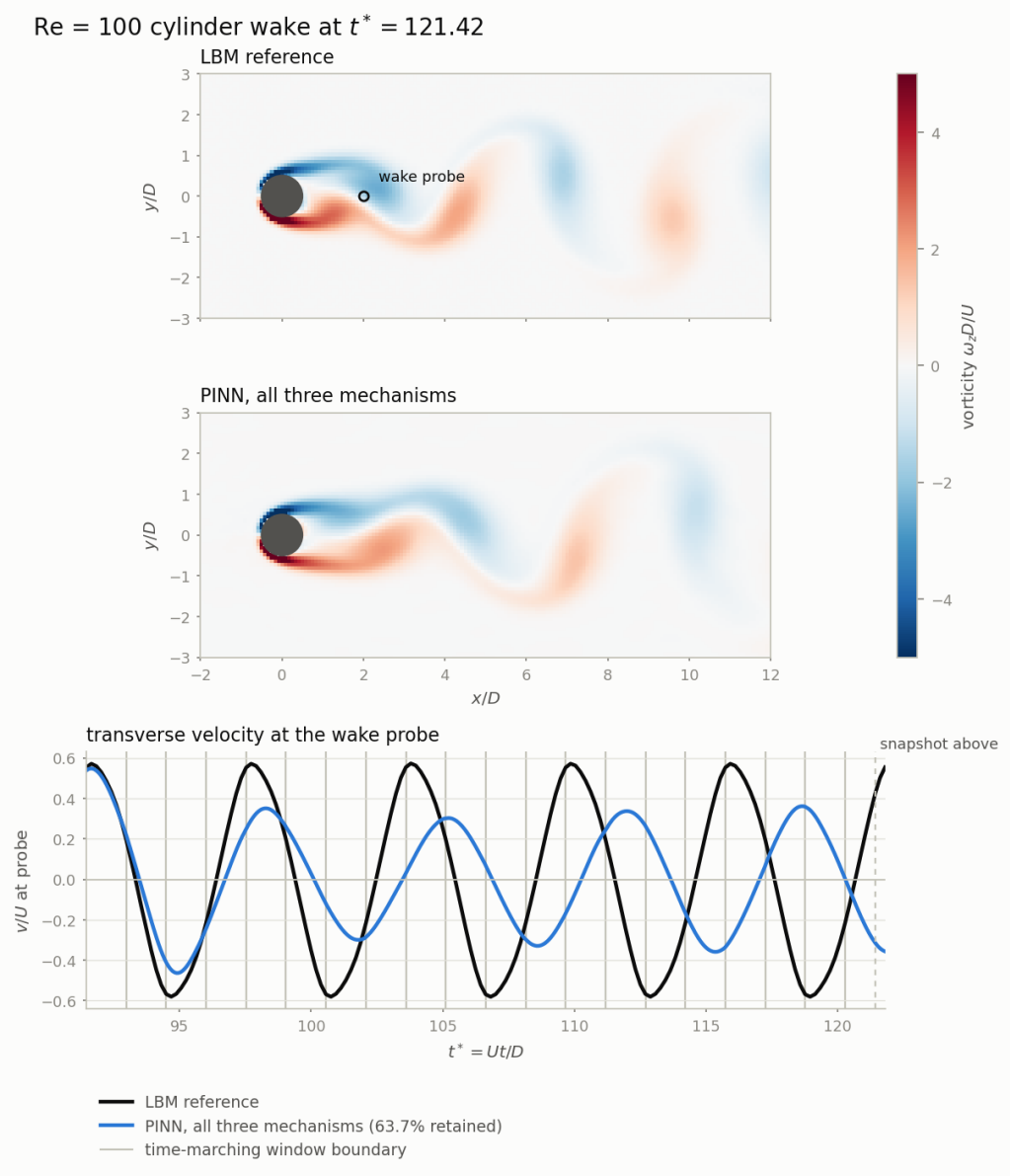


Figure 3: Data-free `all` marched over five periods against the reference. Top: vorticity at the end of the run. Bottom: transverse velocity at the wake probe; grey lines are window boundaries.

4.4 Reynolds-number dependence

The comparison was repeated at Re = 60–180, with new reference data at each Reynolds number; the upper limit is set by the three-dimensional transition at $\mathrm{Re}_c = 188.5$ [11]. Every dataset's St is within 0.6% of Qu et al. The baseline retains 24–45% and `all` 63–79%, and `all` reduces the field error from about 0.17 to 0.028–0.040 (Table 5). These runs use two seeds and a single start phase, the beginning of each record, so the variation in the baseline column reflects start phase and has no physical meaning. The insensitivity of `all` to start phase was measured only at Re = 100.

4.5 Sequence architectures

PINNsFormer [12] and a GRU variant with the same embedding were compared with the MLP. The GRU and MLP were run with two seeds at all four start phases. On a single window the GRU retains 21.8% against the MLP's 30.1%, and marched it retains 33.6% against 56.7%.

Table 5: Data-free results across Reynolds number. Shedding retained for two seeds; L_2 is the two-seed mean.

Re	St vs Qu et al.	baseline	`all`	L_2 baseline	L_2 `all`
60	+0.51%	29.8, 33.5%	62.8, 69.1%	0.163	0.028
80	+0.60%	40.2, 44.6%	75.5, 76.9%	0.173	0.028
100	−0.04%	38.8, 41.5%	71.8, 78.5%	0.173	0.031
120	+0.03%	23.9, 24.1%	71.5, 78.6%	0.170	0.036
150	−0.28%	38.0, 41.4%	69.7, 75.1%	0.174	0.039
180	−0.45%	24.1, 27.7%	77.5, 76.6%	0.171	0.040

PINNsFormer was run once, at phase 0, because of its cost: on a single window it retains 34.9% against the MLP's 40.4%, and marched it reaches 77.5% against 69.7%, at 11.8 times the MLP's run time. Neither architecture prevents the collapse without marching, and one marched run is not enough to show that attention adds to it.

4.6 Physics-informed operator forecasting

Following Renn et al. [5], an FNO maps two consecutive snapshots to the next and is rolled out recursively. It trains on the first 70% of the record and forecasts from states in the last 30%, so this is true forecasting. A physics term is added through the vorticity-transport and continuity residuals, which do not require pressure. On the validated dataset the data-driven FNO reaches an error of 0.0010 at its trained horizon of ten steps and 0.0152 after 50 steps, and the physics term makes it 16–18% worse at ten and twenty steps. Test states on this clean limit cycle are close to the training data, which may explain why physics does not help. On an earlier dataset whose fields contain the acoustic mode, and whose held-out states are therefore less like the training data, physics lowers the FNO error by 21% at ten steps and 28% at twenty, then crosses over at 0.8–1.0 shedding periods. Table 6 repeats this test for a U-Net [13] and a convolutional GRU [14]. The physics term changes the U-Net error by at most 7% and makes the ConvGRU worse, so the short-horizon benefit is specific to the FNO. Without physics, both convolutional networks forecast better than the FNO, with 31 and 131 times fewer parameters.

Table 6: Forecast error on the earlier dataset by number of steps, and the change when physics is added (negative is better). Two seeds.

		data-driven error			change with physics		
backbone	parameters	10	20	50	10	20	50
FNO	59.0 M	0.093	0.183	0.365	−21%	−28%	+111%
U-Net	1.93 M	0.056	0.084	0.159	+1%	−1%	−7%
ConvGRU	0.45 M	0.042	0.092	0.220	+30%	+42%	+22%

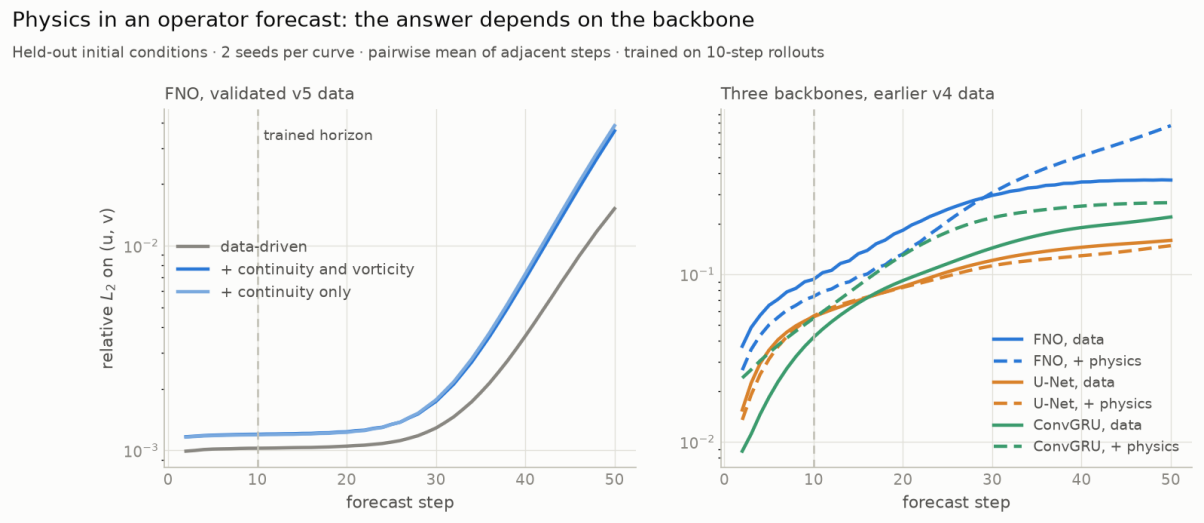


Figure 4: Forecast error against horizon. Left: FNO on the validated dataset. Right: three backbones on the earlier dataset, data-driven (solid) and with physics (dashed).

5 Conclusions

A data-free PINN retains 31.1% of the cylinder wake at Re = 100, and the amount depends on the start phase. No single strategy fixes this. Causal time-marching adds only 7.7 points over an equal-budget baseline, but combined with Fourier features and a Reynolds decomposition it retains 72.9% on average and between 70% and 77% at every start phase, at a field error of 0.031, with a similar gain across Re = 60–180. From physics alone, this PINN predicts about one shedding period; over five it drifts in phase. About 350 supervised points per period from inside the predicted span hold the phase, which is reconstruction rather than forecasting. Adding physics to an operator forecast helped only the FNO, only at short horizons, and only on the earlier dataset.

The main limitations are that most secondary studies use two seeds and PINNsFormer was run once, that continuity between marching windows is enforced only through the loss (exact enforcement [15] is a possible next step), and that the backbone comparison was run only on the earlier dataset.

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